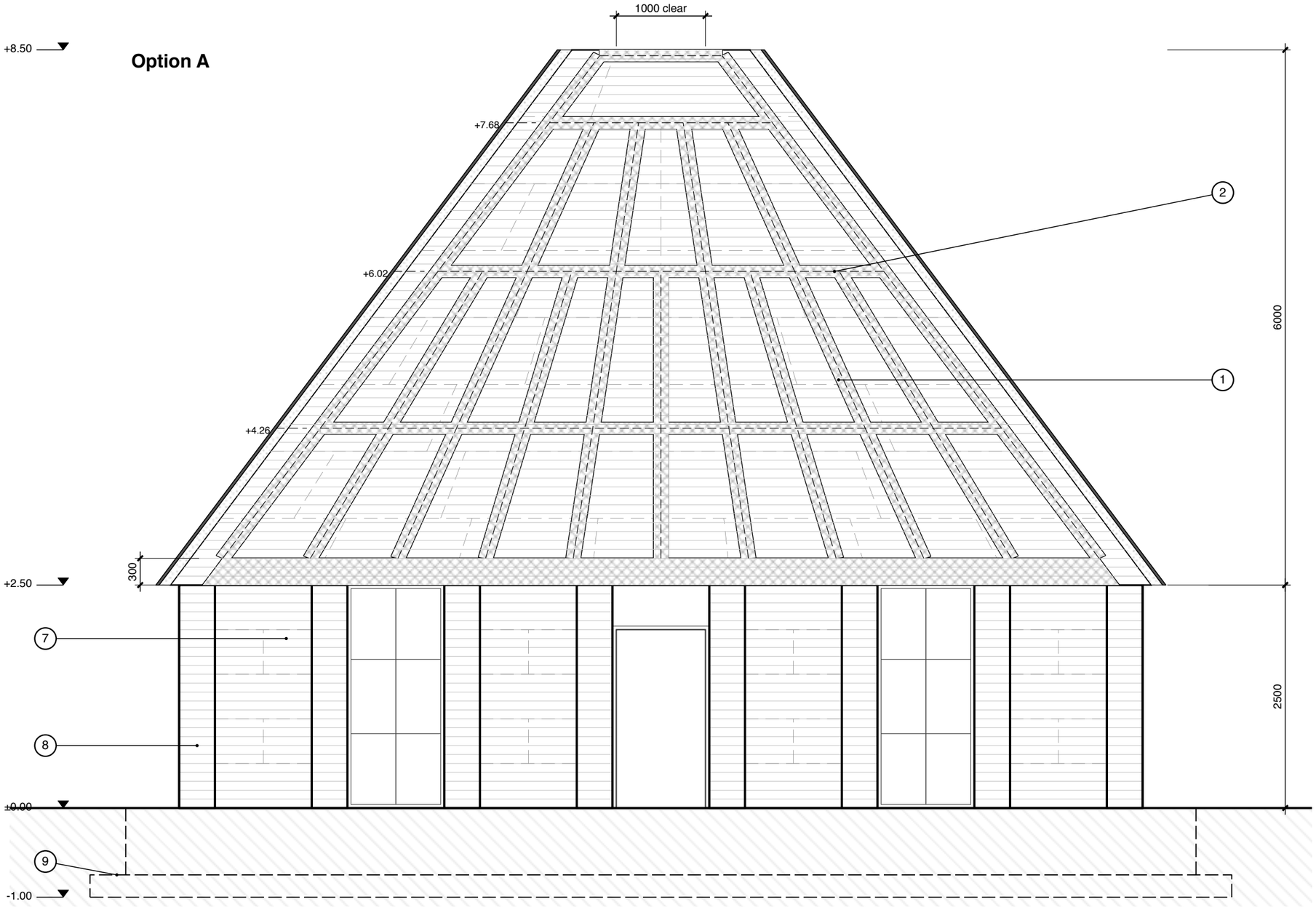
Option A



Roof grid A: ruled-surface columns

one roof face seen square on

1. Columns run in straight lines from the base edge (every 1028) towards the crown edge, so they converge like the ruled surface of the sketch.
2. When the spacing drops to 500 (about +6.02) every other column stops; at about +7.68 only the four hip columns go on to the crown ring.
3. Rib (column) 160 wide x 160 deep in the rib zone, between the inner and the middle bead: the printed channel stays as formwork, filled with cast concrete; 4 Ø16 + Ø8 stirrups every 200 (assumed), or fibre concrete (see notes).
4. Inner bead 30: continuous, covers the grid on the room side.
5. 2 outer beads 30: continuous, the outside of the printed shell (insulation and cladding on them).
6. Lattice zone 120: wavy printed bead 30 touching the middle and the outer beads in turn, insulation between (dotted).
7. Printed wall 400, to +2.50: inner bead 30, lattice zone 310 (wavy bead + insulation), 2 outer beads 30.
8. Buttresses 400 wide, 1000 from the inner face (600 outside), about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included); every other bay is printed wall, the bays between are full-height curtain wall (door in the middle one).
9. Strip footing 1800 x 250, underside at -1.00 (hidden, dashed): printed edge formwork, cast core.
10. Printed channel walls 30 each side of the rib, from the inner to the middle bead: permanent formwork. Between the ribs the rib zone is an empty cavity.
11. Poured core: the solid base (300, full concrete over the whole 400) and the rib in one pour, lifts of about 600 from the eave (assumed, dashed), no joint between base and rib. Printed layer joints every 10 (drawn every 20).
12. Outside the printed shell (seen at the sides of the roof): insulation 100 (type to be chosen), then the outer layer (cladding 30).
13. Middle bead 30: closes the rib zone outside; inner and middle bead are the structural skins, the ribs the webs (as in the Karamba model).

Notes

Columns stop on a horizontal rib (at 3517 and 5183 above the eave); bars are lapped / hooked into it.

Horizontal ribs at 1758, 3517, 5183, plus the 300 solid base (bottom ring) and the top ring at the crown.

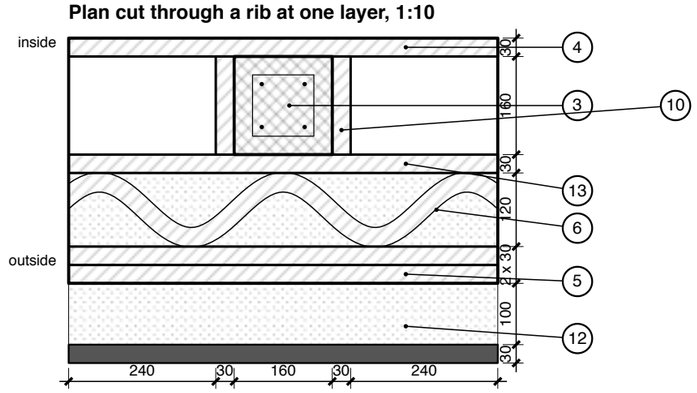
Dashed lines = printed pieces in running bond (about 1500 x 750); very faint lines = printed layer joints (10 mm layers, drawn every 100).

Printed mortar beads are permanent formwork; ribs, base and buttress cores are poured concrete (after the Obayashi 3dpod paper, 2023). Steel-fibre concrete could replace the rib bars, to be confirmed by the engineer.

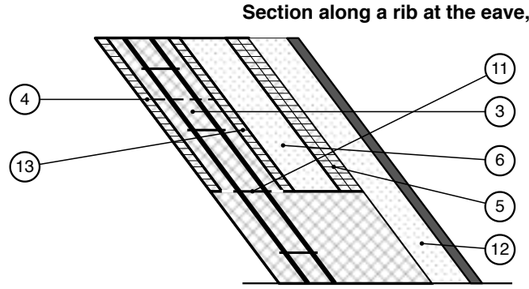
Rough quantities, four faces: ribs 318 m, steel 2005 kg (4 Ø16 per rib), poured concrete 8.1 m3.

0 1 2 3 4 m

Plan cut through a rib at one layer, 1:10



Section along a rib at the eave, 1:20



Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · rib plan cut 1:10, rib section 1:20 · dimensions in mm

2026-10-14 · hand estimates, not engineered

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